

Performance Testing

Date	2-10-2026
Team ID	SWTID-2026-1309
Project Name	Pocket Smart AI
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	k6 (planned)
Testing Tool Used	Load Testing with baseline load and controlled peak load
Type of Testing	
Target Module / API	
Test Environment	Authentication, expense tracking, dashboard and AI insight API flows
Test Date	Local development environment / test server
	2-10-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User login and session creation	10	60	Login requests complete without errors.
2	Expense entry and retrieval	20	120	Expense operations remain responsive under load.
3	Dashboard and report loading	30	120	Dashboard loads within the defined response target.
4	AI insight request and notification flow	20	120	Requests complete successfully with acceptable resource

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	Pending execution	Pending	Measure after baseline run
2	Response Time (Max)	< 5 seconds	Pending execution	Pending	Measure peak response
3	Throughput (Req/sec)		Pending execution	Pending	Record during load test
4	Error Rate	< 1%	Pending execution	Pending	Record failed requests
5	CPU Utilization	< 80%	Pending execution	Pending	Record server CPU usage
6	Memory Utilization	< 80%	Pending execution	Pending	Record server memory usage

Step 4: Observations & Analysis

Key Findings:

No measured performance results were supplied with the project material. Baseline and peak-load execution is pending.

Bottlenecks Identified:

To be identified after the load test; monitor API latency, database access, and AI/API dependency response time.

Optimization Steps Taken:

Run a baseline test first, then tune slow endpoints, database queries, caching, and external API calls based on measured bottlenecks.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Attach k6 result / response-time evidence

Attach throughput / CPU / memory evidence